Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Rafael Pacheco Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
22884 W Ingomar Grade RD	Los Banos	Merced	93635

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1935

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X081-X060-X060-XXXX

B. OPERATORS

Pacheco, Paul

Operator name: Pacheco, Paul

Telephone no.: (209) 827-6877 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
22890 Ingomar Grade RD	Los Banos	CA	95635

C. OWNERS

Pacheco, Rafael and/or Emily

Legal owner name: Pacheco, Rafael and/or Emily

Telephone no.: (209) 827-6611 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
22884 W Ingomar Grade RD	Los Banos	CA	93635

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	178	215	210	105	94
Number under roof	787	0	0	0	0	0
Maximum number	787	178	215	210	105	94
Average number	787	178	215	210	105	94
Avg live weight (lbs)	1,400	1,450	1,100	750		

Predominant milk cow breed: Holstein

Average milk production: 73 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 28,098.15 *tons per reporting period*

Total nitrogen from manure: 350,553.40 *lbs per reporting period*

After ammonia losses (30% loss applied): 245,387.38 *lbs per reporting period*

Total phosphorus from manure: 57,182.91 *lbs per reporting period*

Total potassium from manure: 152,268.94 *lbs per reporting period*

Total salt from manure: 411,490.05 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 12,366,000 *gallons*

Total nitrogen generated: 67,415.11 *lbs*

Total phosphorus generated: 9,489.36 *lbs*

Total potassium generated: 100,618.82 *lbs*

Total salt generated: 525,335.94 *lbs*

	12,366,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	12,366,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - Well A House	Ground water
Domestic Well - Well B House	Ground water
Irrigation water - canal	Surface water
Irrigation water - Gear Head well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Duni 1	35	35	2	process wastewater	X081-X060-X079-XXXX
Duni 2	40	40	2	process wastewater	X081-X060-X079-XXXX
Duni 3	22	22	2	manure	X081-X060-X077-XXXX
Duni 4	68	68	2	manure	X081-X040-X006-XXXX
Duni 5	22	22	0	none	X081-X040-X001-XXXX
Field 1	6	6	2	process wastewater	X081-X060-X060-XXXX
Field 2	20	20	2	process wastewater	X081-X060-X060-XXXX
Field 3	43	43	2	process wastewater	X081-X060-X060-XXXX
Field 4	128	128	2	manure	X081-X060-X060-XXXX
Roselli 1	52	52	0	none	X081-X060-X025-XXXX
Totals for areas that were used for application	362	362	16		
Totals for areas that were not used for application	74	74	0		
Land application area totals	436	436	16		

B. CROPS AND HARVESTS

Duni 1

Field name: Duni 1

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	560.00 *ton*	Dry-weight		71.3	12,500.00	2,900.00	18,300.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.00	114.80	26.63	168.07	900.03

Duni 1

06/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	1,141.00 *ton*	Dry-weight		67.9	12,500.00	2,100.00	18,100.00		5.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.60	261.62	43.95	378.82	1,151.11

Duni 2

Field name: Duni 2

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	640.00 *ton*	Dry-weight		71.8	12,700.00	2,700.00	17,800.00		9.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.00	114.60	24.36	160.63	893.38

06/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	1,302.00 *ton*	Dry-weight		68.0	14,600.00	3,600.00	23,400.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.55	304.15	75.00	487.47	1,229.09

Duni 3

Field name: Duni 3

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 22 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	347.00 *ton*	Dry-weight		70.4	12,400.00	3,000.00	18,000.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.77	115.78	28.01	168.07	896.40

06/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 22 Plant date: 06/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	710.00 *ton*	Dry-weight		68.0	14,000.00	3,200.00	24,000.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.27	289.16	66.09	495.71	1,280.58

Duni 4

Field name: Duni 4

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 68 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	1,074.00 *ton*	Dry-weight		71.9	13,300.00	2,800.00	18,700.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.79	118.05	24.85	165.99	869.88

Duni 4

06/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 68 Plant date: 06/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	2,213.00 *ton*	Dry-weight		67.6	15,700.00	2,900.00	19,800.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.54	331.09	61.16	417.55	1,370.76

Field 1

Field name: Field 1

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 6 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	85.00 *ton*	Dry-weight		72.9	12,900.00	2,700.00	18,600.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.17	99.05	20.73	142.82	767.83

06/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 6 Plant date: 06/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	193.00 *ton*	Dry-weight		67.4	15,900.00	2,700.00	20,300.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.17	333.47	56.63	425.75	1,384.20

Field 2

Field name: Field 2

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	308.00 *ton*	Dry-weight		72.9	12,900.00	2,700.00	18,600.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.40	107.67	22.54	155.25	834.68

06/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	649.00 *ton*	Dry-weight		67.4	15,900.00	2,700.00	20,300.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.45	336.40	57.12	429.50	1,396.39

Field 3

Field name: Field 3

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 43 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	706.00 *ton*	Dry-weight		72.5	13,400.00	2,800.00	18,500.00		9.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.42	121.01	25.28	167.06	875.93

Field 3

06/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 43 Plant date: 06/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	1,394.00 *ton*	Dry-weight		67.6	14,100.00	2,200.00	22,900.00		7.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.42	296.20	46.22	481.07	1,470.51

Field 4

Field name: Field 4

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 128 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	2,337.00 *ton*	Dry-weight		71.8	12,000.00	3,000.00	19,100.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.26	123.57	30.89	196.68	1,009.15

06/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 128 Plant date: 06/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	4,189.00 *ton*	Dry-weight		68.4	14,200.00	2,300.00	15,200.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.73	293.70	47.57	314.38	1,178.94

NUTRIENT BUDGET

A. LAND APPLICATIONS

Duni 1 - 11/08/2022: Oats, silage-soft dough

Field name: Duni 1

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.78	11.02	135.96	723.87	660,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	257.07	5,828,020.00 *gal*
Application event totals		77.78	11.02	135.96	980.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	78.49	11.12	137.20	730.45	666,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	266.97	6,052,520.00 *gal*
Application event totals		78.49	11.12	137.20	997.42	

Duni 1 - 06/23/2023: Corn, silage

Field name: Duni 1

Crop: Corn, silage

Plant date: 06/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	62.93	8.91	110.01	585.68	534,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	257.86	5,845,980.00 *gal*
Application event totals		62.93	8.91	110.01	843.54	

Duni 1 - 06/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	263.01	5,962,720.00 *gal*
Application event totals		0.00	0.00	0.00	263.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	60.81	8.61	106.30	565.93	516,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	263.01	5,962,720.00 *gal*
Application event totals		60.81	8.61	106.30	828.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	62.23	8.81	108.77	579.10	528,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	263.80	5,980,680.00 *gal*
Application event totals		62.23	8.81	108.77	842.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	72.13	10.21	126.07	671.22	612,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	263.41	5,971,700.00 *gal*
Application event totals		72.13	10.21	126.07	934.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	266.97	6,052,520.00 *gal*
Application event totals		0.00	0.00	0.00	266.97	

Duni 2 - 11/08/2022: Oats, silage-soft dough

Field name: Duni 2

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Duni 2 - 11/08/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	75.49	10.69	131.94	702.48	732,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	238.11	6,169,260.00 *gal*
Application event totals		75.49	10.69	131.94	940.59	

03/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	74.87	10.60	130.86	696.72	726,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	251.28	6,510,500.00 *gal*
Application event totals		74.87	10.60	130.86	948.00	

Duni 2 - 06/23/2023: Corn, silage

Field name: Duni 2

Crop: Corn, silage

Plant date: 06/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	113.27	16.19	147.21	717.63	588,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	255.78	6,627,240.00 *gal*
Application event totals		113.27	16.19	147.21	973.41	

06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	264.45	6,851,740.00 *gal*
Application event totals		0.00	0.00	0.00	264.45	

Duni 2 - 06/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	112.12	16.03	145.70	710.31	582,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	262.71	6,806,840.00 *gal*
Application event totals		112.12	16.03	145.70	973.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	59.40	8.41	103.83	552.77	576,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	263.06	6,815,820.00 *gal*
Application event totals		59.40	8.41	103.83	815.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	63.11	8.94	110.31	587.32	612,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	263.41	6,824,800.00 *gal*
Application event totals		63.11	8.94	110.31	850.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	267.91	6,941,540.00 *gal*
Application event totals		0.00	0.00	0.00	267.91	

Duni 3 - 11/08/2022: Oats, silage-soft dough

Field name: Duni 3

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Duni 3 - 11/08/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	243.87	3,475,260.00 *gal*
Application event totals		0.00	0.00	0.00	243.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	250.80	3,574,040.00 *gal*
Application event totals		0.00	0.00	0.00	250.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	147.23	44.92	134.75	0.00	100.00 *ton*
Application event totals		147.23	44.92	134.75	0.00	

Duni 3 - 06/23/2023: Corn, silage

Field name: Duni 3

Crop: Corn, silage

Plant date: 06/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	322.99	74.30	605.03	0.00	200.00 *ton*
Application event totals		322.99	74.30	605.03	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	238.20	3,394,440.00 *gal*
Application event totals		0.00	0.00	0.00	238.20	

Duni 3 - 06/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	245.13	3,493,220.00 *gal*
Application event totals		0.00	0.00	0.00	245.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	247.02	3,520,160.00 *gal*
Application event totals		0.00	0.00	0.00	247.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	245.76	3,502,200.00 *gal*
Application event totals		0.00	0.00	0.00	245.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	246.39	3,511,180.00 *gal*
Application event totals		0.00	0.00	0.00	246.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	252.69	3,600,980.00 *gal*
Application event totals		0.00	0.00	0.00	252.69	

Duni 4 - 11/08/2022: Oats, silage-soft dough

Field name: Duni 4

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Duni 4 - 11/08/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	238.33	10,497,620.00 *gal*
Application event totals		0.00	0.00	0.00	238.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	245.87	10,829,880.00 *gal*
Application event totals		0.00	0.00	0.00	245.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	142.90	43.60	130.79	0.00	300.00 *ton*
Application event totals		142.90	43.60	130.79	0.00	

Duni 4 - 06/23/2023: Corn, silage

Field name: Duni 4

Crop: Corn, silage

Plant date: 06/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	355.28	81.73	665.53	0.00	680.00 *ton*
Application event totals		355.28	81.73	665.53	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	245.67	10,820,900.00 *gal*
Application event totals		0.00	0.00	0.00	245.67	

Duni 4 - 06/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	253.62	11,171,120.00 *gal*
Application event totals		0.00	0.00	0.00	253.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	253.21	11,153,160.00 *gal*
Application event totals		0.00	0.00	0.00	253.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	255.86	11,269,900.00 *gal*
Application event totals		0.00	0.00	0.00	255.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	253.01	11,144,180.00 *gal*
Application event totals		0.00	0.00	0.00	253.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	255.86	11,269,900.00 *gal*
Application event totals		0.00	0.00	0.00	255.86	

Field 1 - 11/08/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 1 - 11/08/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	58.23	7.35	81.01	515.72	72,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	281.89	1,095,560.00 *gal*
Application event totals		58.23	7.35	81.01	797.61	

03/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	56.37	7.05	81.81	424.59	72,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	279.58	1,086,580.00 *gal*
Application event totals		56.37	7.05	81.81	704.18	

Field 1 - 06/23/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage

Plant date: 06/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	146.41	20.93	190.27	927.55	114,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	335.04	1,302,100.00 *gal*
Application event totals		146.41	20.93	190.27	1,262.58	

06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	279.58	1,086,580.00 *gal*
Application event totals		0.00	0.00	0.00	279.58	

Field 1 - 06/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	69.35	9.91	90.13	439.36	54,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	279.58	1,086,580.00 *gal*
Application event totals		69.35	9.91	90.13	718.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	69.35	9.91	90.13	439.36	54,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	281.89	1,095,560.00 *gal*
Application event totals		69.35	9.91	90.13	721.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	69.35	9.91	90.13	439.36	54,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	281.89	1,095,560.00 *gal*
Application event totals		69.35	9.91	90.13	721.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	279.58	1,086,580.00 *gal*
Application event totals		0.00	0.00	0.00	279.58	

Field 2 - 11/08/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 2 - 11/08/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	61.87	8.76	108.15	575.81	300,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	233.60	3,026,260.00 *gal*
Application event totals		61.87	8.76	108.15	809.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	48.26	6.83	84.36	449.13	234,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	235.68	3,053,200.00 *gal*
Application event totals		48.26	6.83	84.36	684.81	

Field 2 - 06/23/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 06/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.47	13.22	120.17	585.82	240,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	266.87	3,457,300.00 *gal*
Application event totals		92.47	13.22	120.17	852.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	250.24	3,241,780.00 *gal*
Application event totals		0.00	0.00	0.00	250.24	

Field 2 - 06/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	90.16	12.89	117.16	571.17	234,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	257.86	3,340,560.00 *gal*
Application event totals		90.16	12.89	117.16	829.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	87.85	12.56	114.16	556.53	228,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	259.94	3,367,500.00 *gal*
Application event totals		87.85	12.56	114.16	816.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	90.16	12.89	117.16	571.17	234,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	268.26	3,475,260.00 *gal*
Application event totals		90.16	12.89	117.16	839.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	259.25	3,358,520.00 *gal*
Application event totals		0.00	0.00	0.00	259.25	

Field 3 - 11/08/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 3 - 11/08/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	69.07	8.72	96.09	611.67	612,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	242.45	6,752,960.00 *gal*
Application event totals		69.07	8.72	96.09	854.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.39	8.63	95.14	605.67	606,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	248.58	6,923,580.00 *gal*
Application event totals		68.39	8.63	95.14	854.25	

Field 3 - 06/23/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 06/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.69	13.68	124.36	606.25	534,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	263.73	7,345,640.00 *gal*
Application event totals		95.69	13.68	124.36	869.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	256.31	7,139,100.00 *gal*
Application event totals		0.00	0.00	0.00	256.31	

Field 3 - 06/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	83.87	11.99	108.99	531.32	468,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	256.96	7,157,060.00 *gal*
Application event totals		83.87	11.99	108.99	788.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	84.94	12.14	110.39	538.14	474,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	256.64	7,148,080.00 *gal*
Application event totals		84.94	12.14	110.39	794.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	80.64	11.53	104.80	510.89	450,000.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	260.83	7,264,820.00 *gal*
Application event totals		80.64	11.53	104.80	771.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	256.31	7,139,100.00 *gal*
Application event totals		0.00	0.00	0.00	256.31	

Field 4 - 11/08/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 4 - 11/08/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	247.16	20,492,360.00 *gal*
Application event totals		0.00	0.00	0.00	247.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	248.57	20,609,100.00 *gal*
Application event totals		0.00	0.00	0.00	248.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	160.69	49.02	147.07	0.00	635.00 *ton*
Application event totals		160.69	49.02	147.07	0.00	

Field 4 - 06/23/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 06/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	349.73	80.45	655.13	0.00	1,260.00 *ton*
Application event totals		349.73	80.45	655.13	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	264.27	21,911,200.00 *gal*
Application event totals		0.00	0.00	0.00	264.27	

Field 4 - 06/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	251.17	20,824,620.00 *gal*
Application event totals		0.00	0.00	0.00	251.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	248.24	20,582,160.00 *gal*
Application event totals		0.00	0.00	0.00	248.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	249.54	20,689,920.00 *gal*
Application event totals		0.00	0.00	0.00	249.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	257.67	21,363,420.00 *gal*
Application event totals		0.00	0.00	0.00	257.67	

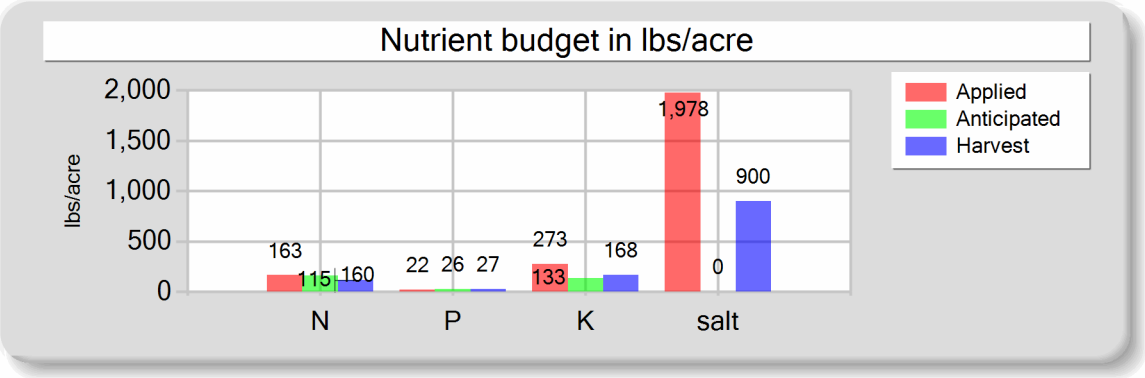
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	264.38	21,920,180.00 *gal*
Application event totals		0.00	0.00	0.00	264.38	

B. NUTRIENT BUDGET

Duni 1 - 11/08/2022: Oats, silage-soft dough

Field name: Duni 1 Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	156.28	22.13	273.16	1,454.32
Fresh water	0.00	0.00	0.00	524.04
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	163.28	22.13	273.16	1,978.36
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	114.80	26.63	168.07	900.03
Nutrient balance	48.48	-4.50	105.09	1,078.33
Applied to removed ratio	1.42	0.83	1.63	2.20

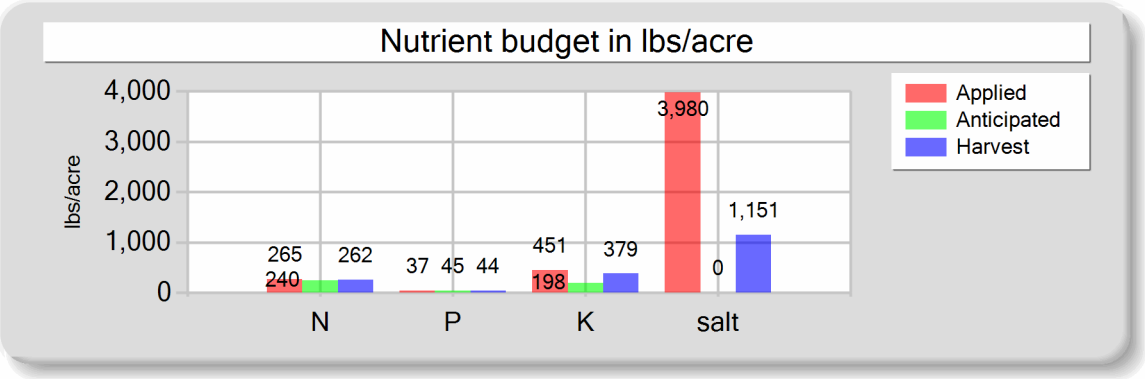
Fresh water applied	
11,880,540.00	*gallons*
437.52	*acre-inches*
12.50	*inches/acre*

Process wastewater applied	
1,326,000.00	*gallons*
48.83	*acre-inches*
1.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

Duni 1 - 06/23/2023: Corn, silage

Field name: Duni 1 Crop: Corn, silage Plant date: 06/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	258.10	36.55	451.15	2,401.93
Fresh water	0.00	0.00	0.00	1,578.07
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	265.10	36.55	451.15	3,980.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	261.62	43.95	378.82	1,151.11
Nutrient balance	3.49	-7.40	72.33	2,828.89
Applied to removed ratio	1.01	0.83	1.19	3.46

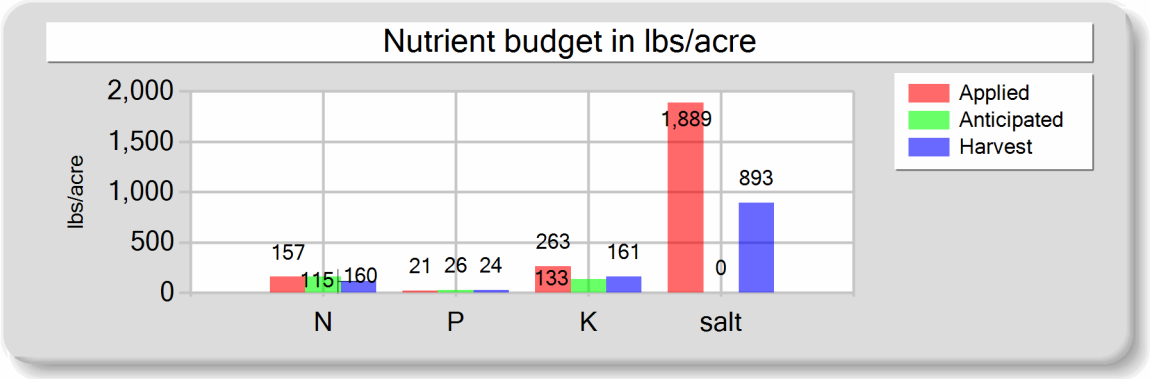
Fresh water applied	
35,776,320.00	*gallons*
1,317.52	*acre-inches*
37.64	*inches/acre*

Process wastewater applied	
2,190,000.00	*gallons*
80.65	*acre-inches*
2.30	*inches/acre*

Total harvests for the crop	
1	*harvests*

Duni 2 - 11/08/2022: Oats, silage-soft dough

Field name: Duni 2 Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	150.35	21.29	262.81	1,399.21
Fresh water	0.00	0.00	0.00	489.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	157.35	21.29	262.81	1,888.59
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	114.60	24.36	160.63	893.38
Nutrient balance	42.75	-3.07	102.18	995.21
Applied to removed ratio	1.37	0.87	1.64	2.11

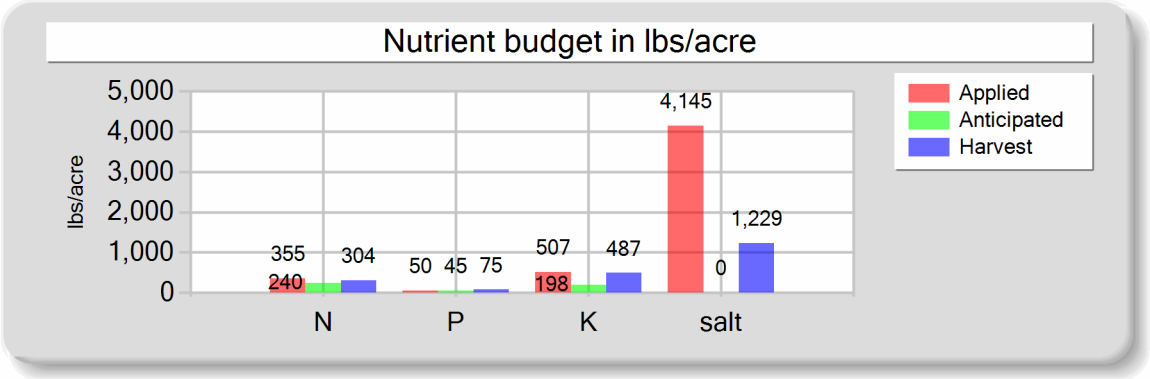
Fresh water applied	
12,679,760.00	*gallons*
466.95	*acre-inches*
11.67	*inches/acre*

Process wastewater applied	
1,458,000.00	*gallons*
53.69	*acre-inches*
1.34	*inches/acre*

Total harvests for the crop	
1	*harvests*

Duni 2 - 06/23/2023: Corn, silage

Field name: Duni 2 Crop: Corn, silage Plant date: 06/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	347.90	49.57	507.05	2,568.03
Fresh water	0.00	0.00	0.00	1,577.33
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	354.90	49.57	507.05	4,145.35
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	304.15	75.00	487.47	1,229.09
Nutrient balance	50.76	-25.43	19.58	2,916.26
Applied to removed ratio	1.17	0.66	1.04	3.37

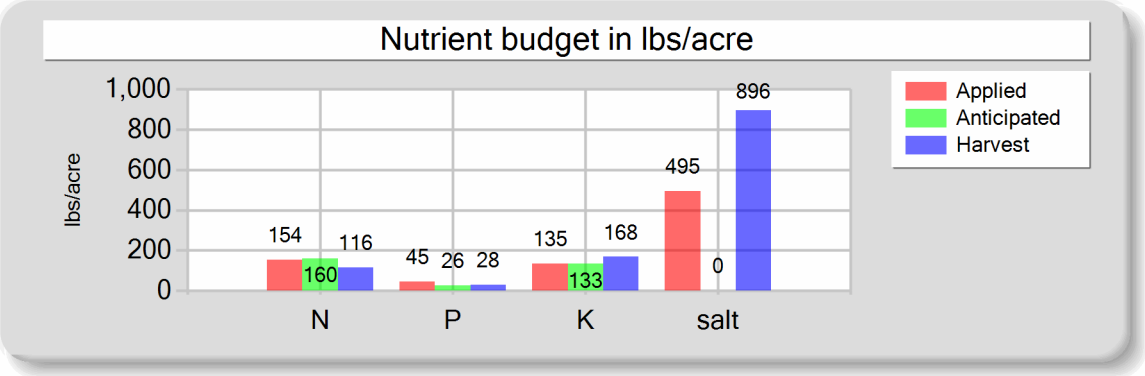
Fresh water applied	
40,867,980.00	*gallons*
1,505.03	*acre-inches*
37.63	*inches/acre*

Process wastewater applied	
2,358,000.00	*gallons*
86.84	*acre-inches*
2.17	*inches/acre*

Total harvests for the crop	
1	*harvests*

Duni 3 - 11/08/2022: Oats, silage-soft dough

Field name: Duni 3 Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	147.23	44.92	134.75	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	494.68
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	154.23	44.92	134.75	494.68
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	115.78	28.01	168.07	896.40
Nutrient balance	38.45	16.91	-33.32	-401.72
Applied to removed ratio	1.33	1.60	0.80	0.55

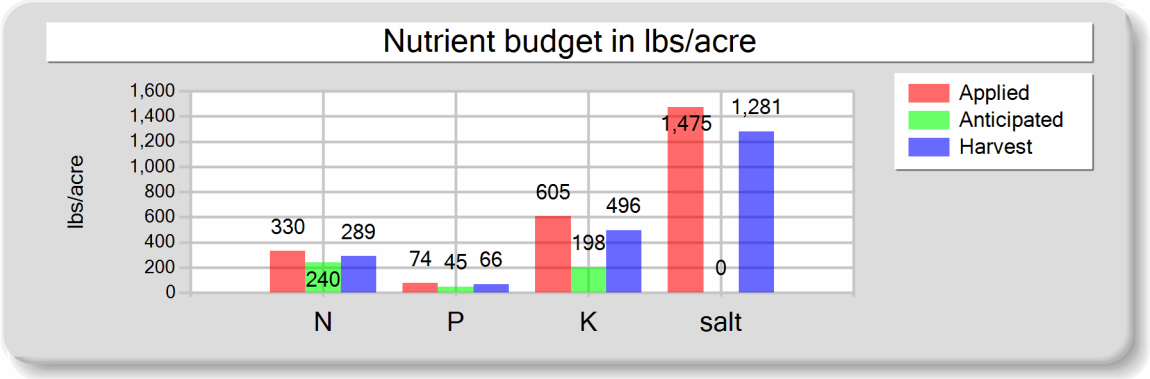
Fresh water applied	
7,049,300.00	*gallons*
259.60	*acre-inches*
11.80	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Duni 3 - 06/23/2023: Corn, silage

Field name: Duni 3 Crop: Corn, silage Plant date: 06/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	322.99	74.30	605.03	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,475.21
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	329.99	74.30	605.03	1,475.21
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	289.16	66.09	495.71	1,280.58
Nutrient balance	40.82	8.21	109.32	194.63
Applied to removed ratio	1.14	1.12	1.22	1.15

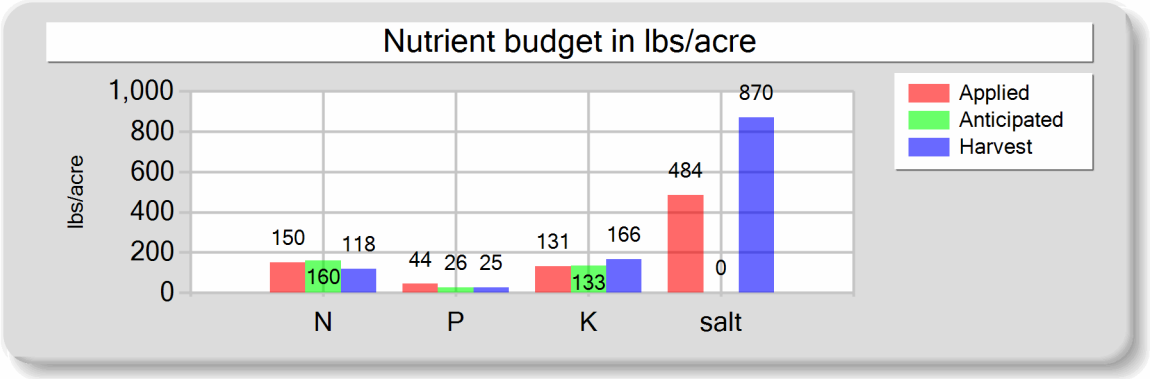
Fresh water applied	
21,022,180.00	*gallons*
774.18	*acre-inches*
35.19	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Duni 4 - 11/08/2022: Oats, silage-soft dough

Field name: Duni 4 Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	142.90	43.60	130.79	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	484.20
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	149.90	43.60	130.79	484.20
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	118.05	24.85	165.99	869.88
Nutrient balance	31.85	18.74	-35.20	-385.67
Applied to removed ratio	1.27	1.75	0.79	0.56

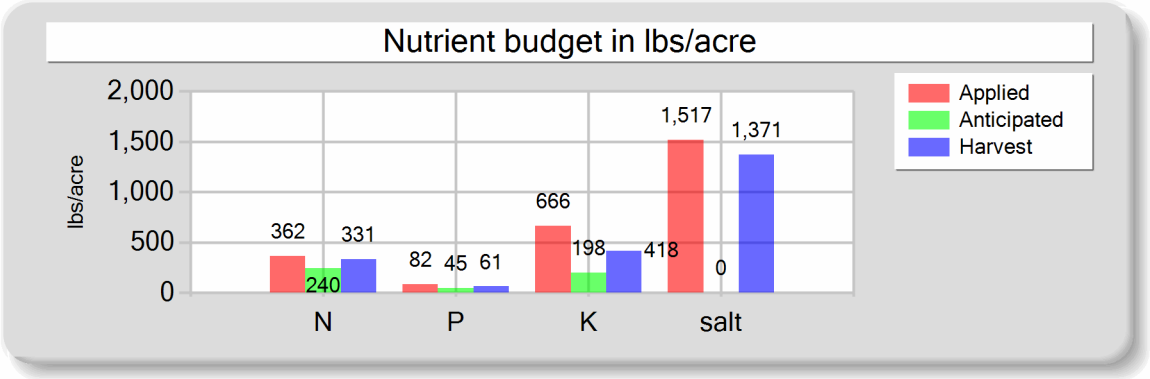
Fresh water applied	
21,327,500.00	*gallons*
785.42	*acre-inches*
11.55	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Duni 4 - 06/23/2023: Corn, silage

Field name: Duni 4 Crop: Corn, silage Plant date: 06/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	355.28	81.73	665.53	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,517.24
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	362.28	81.73	665.53	1,517.24
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	331.09	61.16	417.55	1,370.76
Nutrient balance	31.19	20.58	247.98	146.48
Applied to removed ratio	1.09	1.34	1.59	1.11

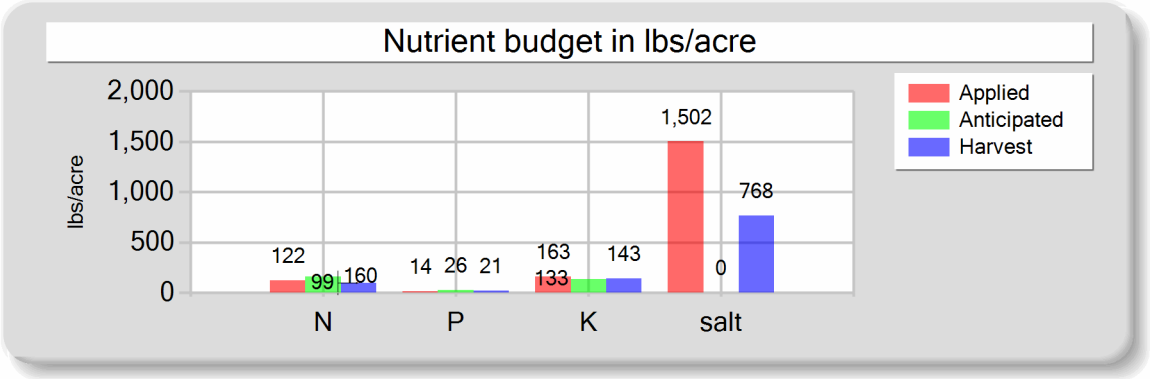
Fresh water applied	
66,829,160.00	*gallons*
2,461.09	*acre-inches*
36.19	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 11/08/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	114.60	14.40	162.83	940.31
Fresh water	0.00	0.00	0.00	561.47
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	121.60	14.40	162.83	1,501.79
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	99.05	20.73	142.82	767.83
Nutrient balance	22.55	-6.33	20.01	733.95
Applied to removed ratio	1.23	0.69	1.14	1.96

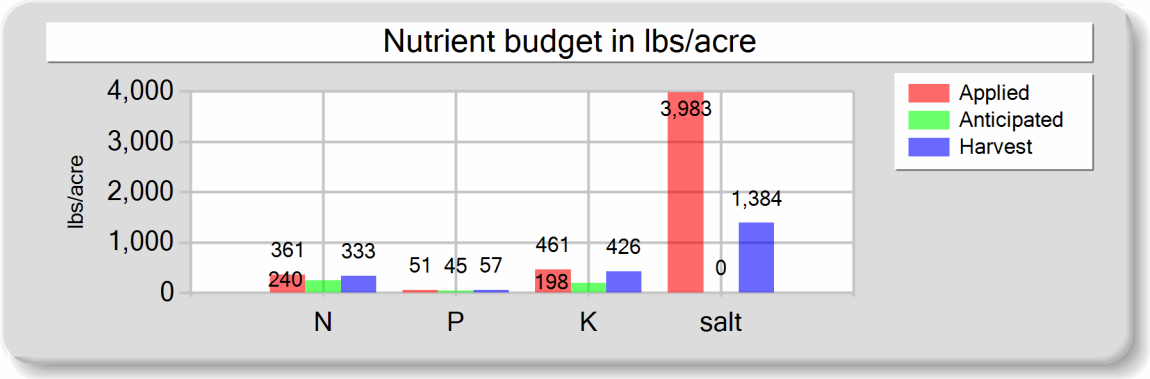
Fresh water applied	
2,182,140.00	*gallons*
80.36	*acre-inches*
13.39	*inches/acre*

Process wastewater applied	
144,000.00	*gallons*
5.30	*acre-inches*
0.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 06/23/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 06/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	354.47	50.67	460.64	2,245.64
Fresh water	0.00	0.00	0.00	1,737.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	361.47	50.67	460.64	3,983.20
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	333.47	56.63	425.75	1,384.20
Nutrient balance	28.00	-5.96	34.90	2,599.01
Applied to removed ratio	1.08	0.89	1.08	2.88

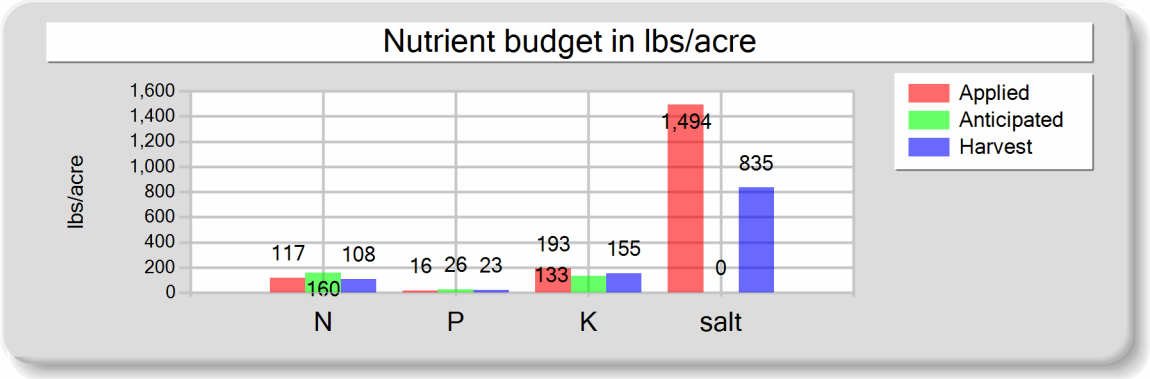
Fresh water applied	
6,752,960.00	*gallons*
248.69	*acre-inches*
41.45	*inches/acre*

Process wastewater applied	
276,000.00	*gallons*
10.16	*acre-inches*
1.69	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 11/08/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	110.14	15.60	192.51	1,024.93
Fresh water	0.00	0.00	0.00	469.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	117.14	15.60	192.51	1,494.21
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	107.67	22.54	155.25	834.68
Nutrient balance	9.46	-6.94	37.26	659.53
Applied to removed ratio	1.09	0.69	1.24	1.79

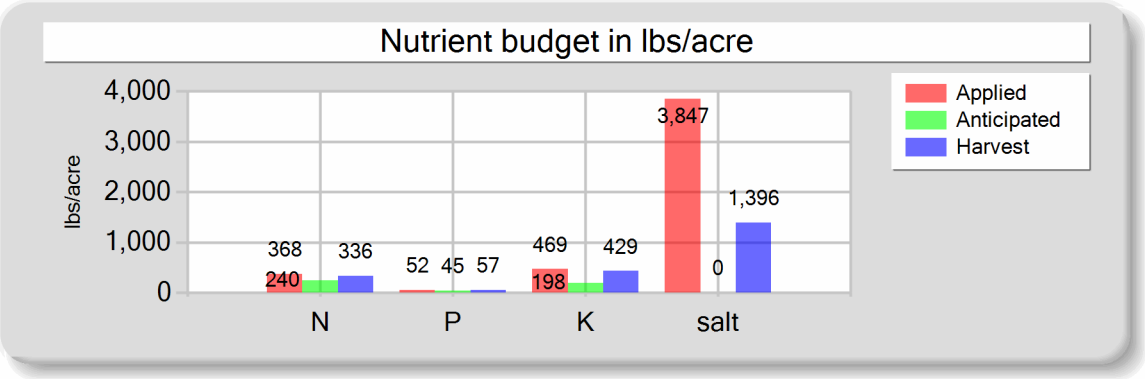
Fresh water applied	
6,079,460.00	*gallons*
223.89	*acre-inches*
11.19	*inches/acre*

Process wastewater applied	
534,000.00	*gallons*
19.67	*acre-inches*
0.98	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 06/23/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 06/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	360.63	51.55	468.66	2,284.69
Fresh water	0.00	0.00	0.00	1,562.42
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	367.63	51.55	468.66	3,847.12
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	336.40	57.12	429.50	1,396.39
Nutrient balance	31.23	-5.57	39.16	2,450.73
Applied to removed ratio	1.09	0.90	1.09	2.76

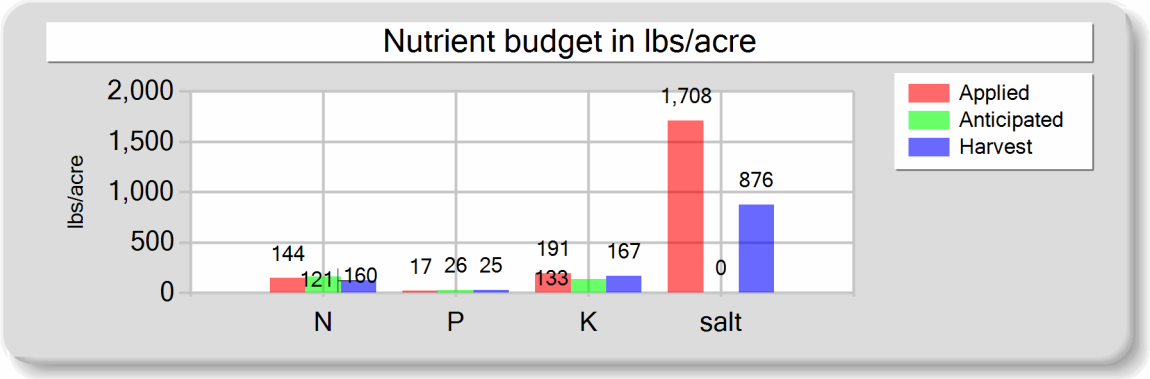
Fresh water applied	
20,240,920.00	*gallons*
745.40	*acre-inches*
37.27	*inches/acre*

Process wastewater applied	
936,000.00	*gallons*
34.47	*acre-inches*
1.72	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 11/08/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	137.45	17.35	191.23	1,217.34
Fresh water	0.00	0.00	0.00	491.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	144.45	17.35	191.23	1,708.37
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	121.01	25.28	167.06	875.93
Nutrient balance	23.45	-7.93	24.17	832.44
Applied to removed ratio	1.19	0.69	1.14	1.95

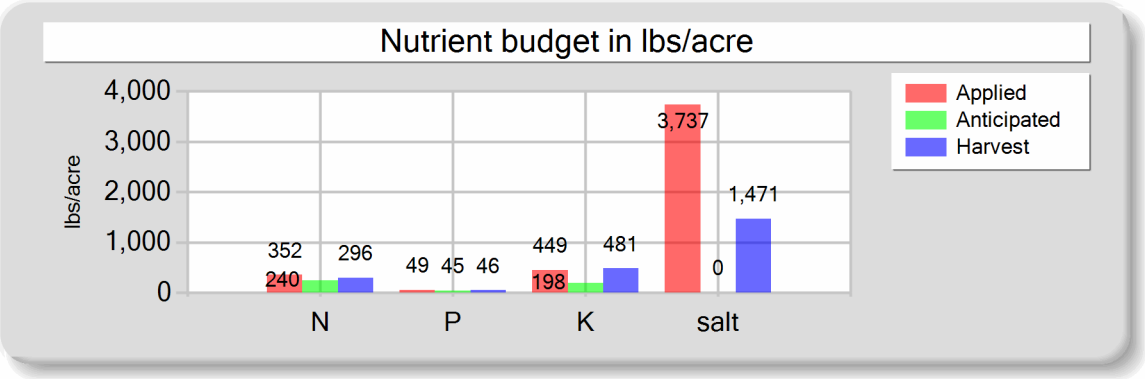
Fresh water applied	
13,676,540.00	*gallons*
503.66	*acre-inches*
11.71	*inches/acre*

Process wastewater applied	
1,218,000.00	*gallons*
44.85	*acre-inches*
1.04	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 06/23/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 06/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	345.15	49.34	448.53	2,186.60
Fresh water	0.00	0.00	0.00	1,550.78
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	352.15	49.34	448.53	3,737.39
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	296.20	46.22	481.07	1,470.51
Nutrient balance	55.94	3.12	-32.53	2,266.88
Applied to removed ratio	1.19	1.07	0.93	2.54

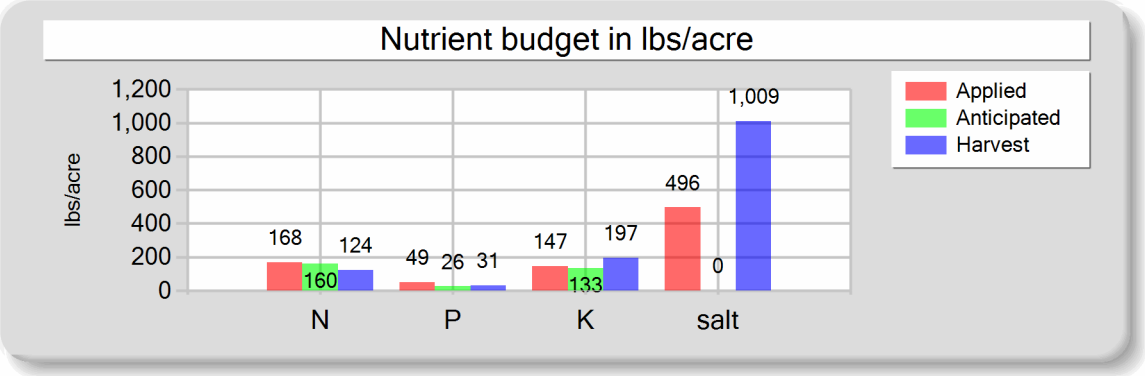
Fresh water applied	
43,193,800.00	*gallons*
1,590.68	*acre-inches*
36.99	*inches/acre*

Process wastewater applied	
1,926,000.00	*gallons*
70.93	*acre-inches*
1.65	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 11/08/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	160.69	49.02	147.07	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	495.73
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	167.69	49.02	147.07	495.73
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	123.57	30.89	196.68	1,009.15
Nutrient balance	44.12	18.13	-49.61	-513.42
Applied to removed ratio	1.36	1.59	0.75	0.49

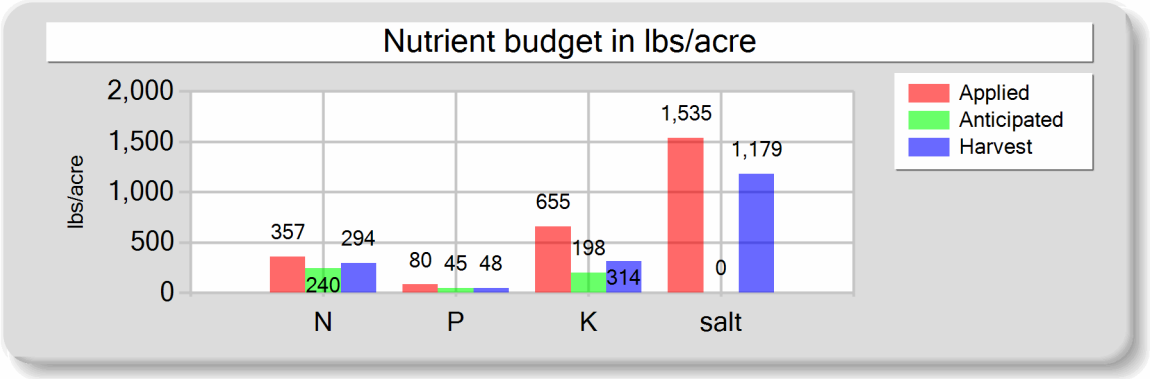
Fresh water applied	
41,101,460.00	*gallons*
1,513.63	*acre-inches*
11.83	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 06/23/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 06/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	349.73	80.45	655.13	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,535.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	356.73	80.45	655.13	1,535.28
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	293.70	47.57	314.38	1,178.94
Nutrient balance	63.03	32.88	340.75	356.34
Applied to removed ratio	1.21	1.69	2.08	1.30

Fresh water applied	
127,291,500.00	*gallons*
4,687.71	*acre-inches*
36.62	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure

Sample date: 05/31/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,700.00	5,400.00	16,200.00							
DL	100.00	100.00	100.00							

Solid Manure

Sample and source description: Solid Manure

Sample date: 10/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 16.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,300.00	4,900.00	39,900.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	922.00	443.00	0.00	1.40	132.00	1,200.00								10,800.00	5,850
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	562.00	411.00	0.00	0.90	70.40	817.00								9,010.00	4,240
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	581.00	479.00	0.00	0.50	73.40	809.00								9,290.00	5,150
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	493.00	450.00	0.00	1.30	70.00	864.00								8,260.00	4,600
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - Well A House

Domestic Well - Well A House

Well A House

Sample description: Well A House

Sample date: 11/08/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			1.20								532.00	
DL			0.10								10.00	

Domestic Well - Well B House

Well B House

Sample description: Well B House

Sample date: 11/08/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			1.10								516.00	
DL			0.10								10.00	

Irrigation water - canal

Irrigation water - canal water

Sample description: Irrigation water - canal water

Sample date: 07/19/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00										277.00	185
DL	1.00										10.00	10

Irrigation water - Gear Head well

Irrigation water - Gear Head well

Irrigation water - Gear Head well

Sample description: Irrigation water - Gear Head well

Sample date: 07/19/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	31.50	0.00	30.70								2,370.00	1,520
DL	1.00	0.50	0.10								10.00	10

D. SOIL ANALYSES

Duni 1

Duni 1

Sample and source description: Duni 1

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			49.00				
DL			2.00				

Duni 2

Duni 2

Sample and source description: Duni 2

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			28.00				
DL			2.00				

Duni 3

Duni 3

Duni 3

Sample and source description: Duni 3

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			108.00				
DL			2.00				

Duni 4

Duni 4

Sample and source description: Duni 4

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			42.00				
DL			2.00				

Field 1

Field 1

Sample and source description: Field 1

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			88.00				
DL			2.00				

Field 2

Field 2

Field 2

Sample and source description: Field 2

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			211.00				
DL			2.00				

Field 3

Field 3

Sample and source description: Field 3

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			105.00				
DL			2.00				

Field 4

Field 4

Sample and source description: Field 4

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			155.00				
DL			2.00				

E. PLANT TISSUE ANALYSES

Duni 1 - 11/08/2022: Oats, silage-soft dough

Duni 1 - 11/08/2022: Oats, silage-soft dough

Duni 1

Sample and source description: Duni 1

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	2,900.00	18,300.00		9.80
DL	100.00	100.00	100.00		0.01

Duni 1 - 06/23/2023: Corn, silage

Duni 1

Sample and source description: Duni 1

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	2,100.00	18,100.00		5.50
DL	100.00	100.00	100.00		0.01

Duni 2 - 11/08/2022: Oats, silage-soft dough

Duni 2

Sample and source description: Duni 2

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,700.00	2,700.00	17,800.00		9.90
DL	100.00	100.00	100.00		0.01

Duni 2 - 06/23/2023: Corn, silage

Duni 2

Sample and source description: Duni 2

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	3,600.00	23,400.00		5.90
DL	100.00	100.00	100.00		0.01

Duni 3 - 11/08/2022: Oats, silage-soft dough

Duni 3

Sample and source description: Duni 3

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	3,000.00	18,000.00		9.60
DL	100.00	100.00	100.00		0.01

Duni 3 - 06/23/2023: Corn, silage

Duni 3

Sample and source description: Duni 3

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,000.00	3,200.00	24,000.00		6.20
DL	100.00	100.00	100.00		0.01

Duni 4 - 11/08/2022: Oats, silage-soft dough

Duni 4 A

Sample and source description: Duni 4 A

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	2,700.00	18,600.00		9.10
DL	100.00	100.00	100.00		0.01

Duni 4 B

Sample and source description: Duni 4 B

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,300.00	2,800.00	18,700.00		9.80
DL	100.00	100.00	100.00		0.01

Duni 4 - 06/23/2023: Corn, silage

Duni 4 A

Sample and source description: Duni 4 A

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,700.00	2,900.00	19,800.00		6.50
DL	100.00	100.00	100.00		0.01

Duni 4 - 06/23/2023: Corn, silage

Duni 4 B

Sample and source description: Duni 4 B

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,700.00	3,000.00	17,900.00		6.20
DL	100.00	100.00	100.00		0.01

Field 1 - 11/08/2022: Oats, silage-soft dough

Field 1 & 2

Sample and source description: Field 1 & 2

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	2,700.00	18,600.00		10.00
DL	100.00	100.00	100.00		0.01

Field 1 - 06/23/2023: Corn, silage

Field 1 & 2

Sample and source description: Field 1 & 2

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,900.00	2,700.00	20,300.00		6.60
DL	100.00	100.00	100.00		0.01

Field 2 - 11/08/2022: Oats, silage-soft dough

Field 1 & 2

Sample and source description: Field 1 & 2

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	2,700.00	18,600.00		10.00
DL	100.00	100.00	100.00		0.01

Field 2 - 06/23/2023: Corn, silage

Field 1 & 2

Sample and source description: Field 1 & 2

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,900.00	2,700.00	20,300.00		6.60
DL	100.00	100.00	100.00		0.01

Field 3 - 11/08/2022: Oats, silage-soft dough

Field 3

Sample and source description: Field 3

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	2,800.00	18,500.00		9.70
DL	100.00	100.00	100.00		0.01

Field 3 - 06/23/2023: Corn, silage

Field 3

Sample and source description: Field 3

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,100.00	2,200.00	22,900.00		7.00
DL	100.00	100.00	100.00		0.01

Field 4 - 11/08/2022: Oats, silage-soft dough

Field 4 A

Sample and source description: Field 4 A

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	2,700.00	18,700.00		9.70
DL	100.00	100.00	100.00		0.01

Field 4 B

Sample and source description: Field 4 B

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	3,000.00	19,100.00		9.80
DL	100.00	100.00	100.00		0.01

Field 4 - 11/08/2022: Oats, silage-soft dough

Field 4 C

Sample and source description: Field 4 C

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	2,900.00	17,800.00		9.10
DL	100.00	100.00	100.00		0.01

Field 4 - 06/23/2023: Corn, silage

Field 4 A

Sample and source description: Field 4 A

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,200.00	2,500.00	15,400.00		5.50
DL	100.00	100.00	100.00		0.01

Field 4 B

Sample and source description: Field 4 B

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,200.00	2,300.00	15,200.00		5.70
DL	100.00	100.00	100.00		0.01

Field 4 - 06/23/2023: Corn, silage

Field 4 C

Sample and source description: Field 4 C

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,600.00	2,900.00	17,100.00		6.10
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

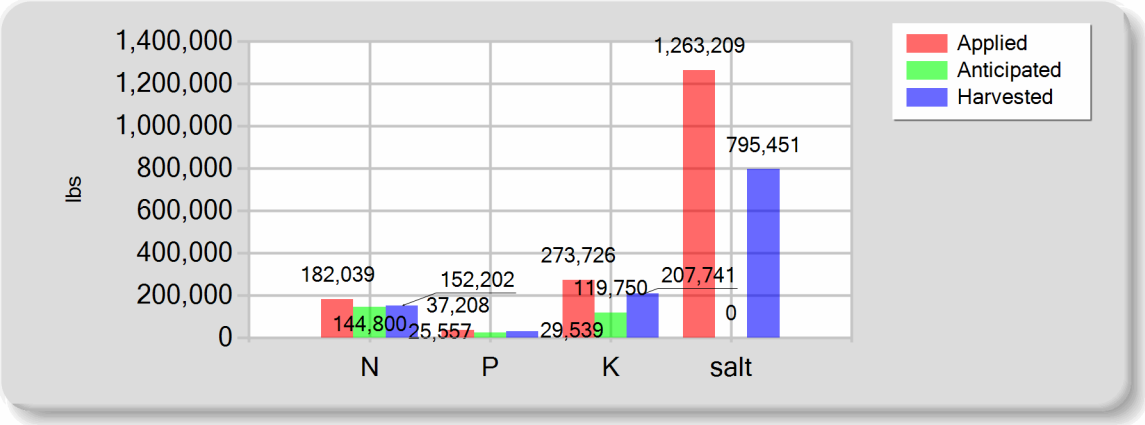
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

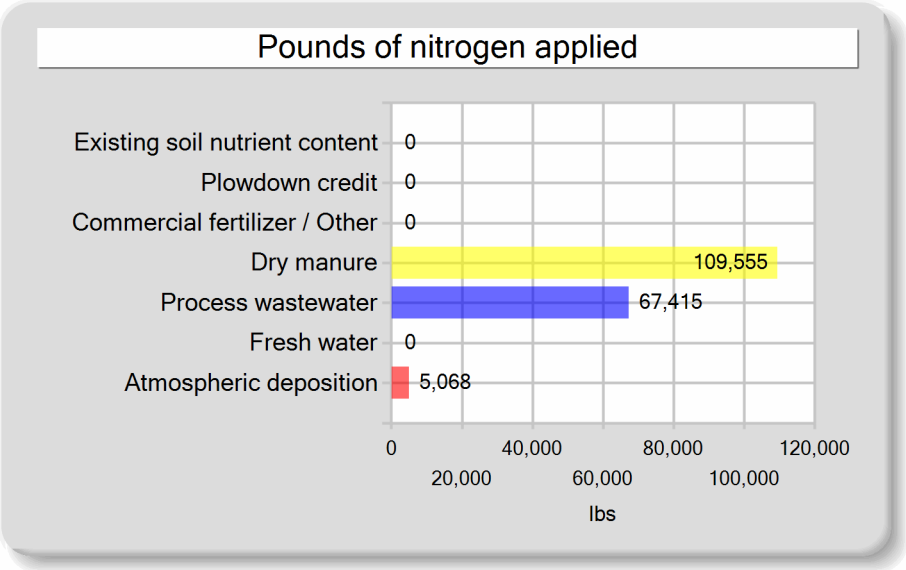
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

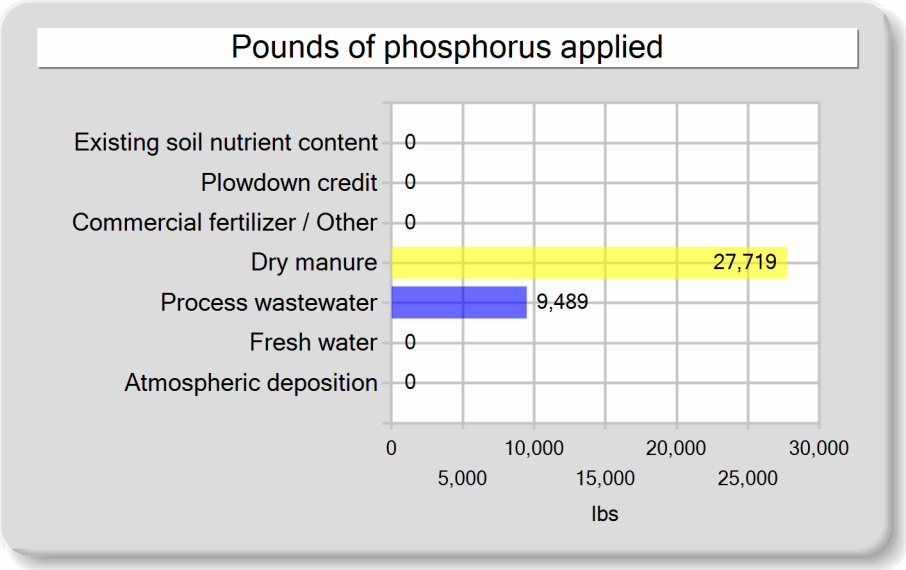
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	109,555.46	27,718.52	173,107.46	0.00
Process wastewater	67,415.11	9,489.36	100,618.82	525,335.94
Fresh water	0.00	0.00	0.00	737,873.51
Atmospheric deposition	5,068.00	0.00	0.00	0.00
Total nutrients applied	182,038.58	37,207.87	273,726.28	1,263,209.45
Anticipated crop nutrient removal	144,800.00	25,557.20	119,749.60	0.00
Actual crop nutrient removal	152,202.48	29,539.17	207,740.83	795,450.76
Nutrient balance	29,836.10	7,668.70	65,985.44	467,758.69
Applied to removed ratio	1.20	1.26	1.32	1.59

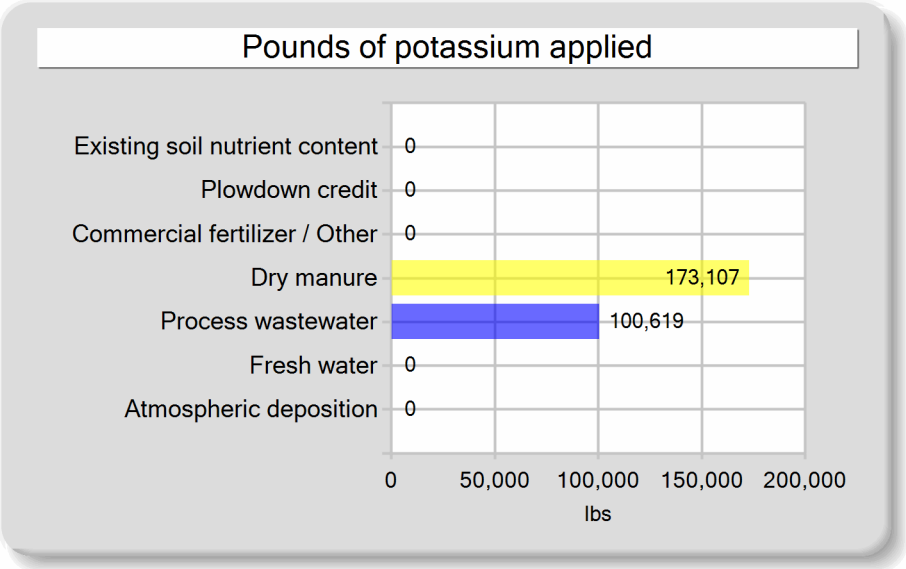
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

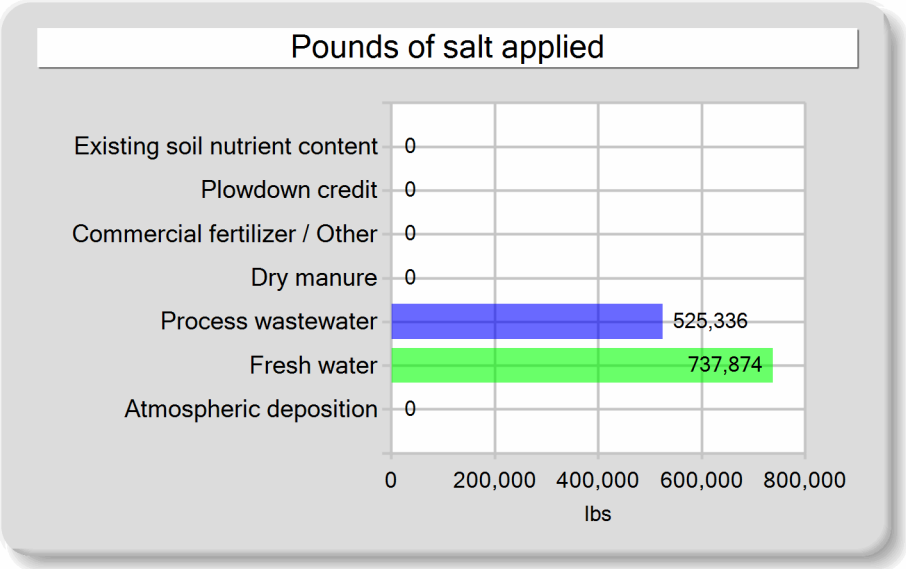


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Rafael Pacheco	Paul Pacheco
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Rafael and/or Emily Pacheco	Paul Pacheco
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-25-2024	6-25-2024
DATE	DATE

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Rafael Pacheco

Received: 07/20/2023 15:31
Reported: 07/31/2023 08:53

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23G1468-01	Gear Head Well	Ag Water	T. Madruga		07/19/2023 16:00

Default Cooler Temperature on Receipt °C: -0.6
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Sample Results

Sample: Gear Head Well
23G1468-01 (Water)

Sampled: 7/19/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**2.37**	mmhos/cm	0.01	1		07/26/23 14:09	SM 2510 B		BEG0838
Electrical Conductivity umhos	**2370**	umhos/cm	10.0	1		07/26/23 14:09	SM 2510 B		BEG0838
Nitrate Nitrogen as NO3N	**30.7**	mg/L	0.1	1	10	07/21/23 06:21	EPA 300.0		BEG0659
pH	**7.6**	units	1.0	1		07/26/23 14:09	SM 4500-H+	H	BEG0838
Total Filterable Solids (TDS)	**1520**	mg/L	10.0	1		07/27/23 15:43	SM 2540 C		BEG0710
Temperature	**25.0**	°C	0.0	1		07/26/23 14:09	SM 2510 B		BEG0838
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		07/24/23 11:26	SM 4500-NH3 C		BEG0809
Total Nitrogen	**31.5**	mg/L	1.00	1		07/24/23 11:26	SM 4500-NH3 C		BEG0809

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Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG0659										
Blank (BEG0659-BLK1)					Prepared & Analyzed: 7/20/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEG0659-BLK2)					Prepared & Analyzed: 7/20/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEG0659-BLK3)					Prepared & Analyzed: 7/21/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEG0659-BLK4)					Prepared & Analyzed: 7/21/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEG0659-BS1)					Prepared & Analyzed: 7/20/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
LCS (BEG0659-BS2)					Prepared & Analyzed: 7/21/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		100	90-110		
LCS (BEG0659-BS3)					Prepared & Analyzed: 7/21/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		100	90-110		
Duplicate (BEG0659-DUP1)		**Source: 23G1426-01**			Prepared & Analyzed: 7/20/2023					
Nitrate Nitrogen as NO3N	0.3		0.1	mg/L		0.3			0.678	10
Duplicate (BEG0659-DUP2)		**Source: 23G1426-06**			Prepared & Analyzed: 7/21/2023					
Nitrate Nitrogen as NO3N	0.2		0.1	mg/L		0.2			0.823	10
Duplicate (BEG0659-DUP3)		**Source: 23G1451-01**			Prepared & Analyzed: 7/21/2023					
Nitrate Nitrogen as NO3N	0.2		0.1	mg/L		0.2			0.411	10
Matrix Spike (BEG0659-MS1)		**Source: 23G1426-01**			Prepared & Analyzed: 7/20/2023					
Nitrate Nitrogen as NO3N	5.4		0.1	mg/L	5.000	0.3	102	90-110		
Matrix Spike (BEG0659-MS2)		**Source: 23G1426-06**			Prepared & Analyzed: 7/21/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L	5.000	0.2	102	90-110		
Matrix Spike (BEG0659-MS3)		**Source: 23G1451-01**			Prepared & Analyzed: 7/21/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L	5.000	0.2	102	90-110		
Reference (BEG0659-SRM1)					Prepared & Analyzed: 7/20/2023					
Nitrate Nitrogen as NO3N	10.0			mg/L	10.00		100	90-110		
Reference (BEG0659-SRM2)					Prepared & Analyzed: 7/20/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG0659 (Continued)										
Reference (BEG0659-SRM3)					Prepared & Analyzed: 7/21/2023					
Nitrate Nitrogen as NO3N	10.4			mg/L	10.00		104	90-110		
Reference (BEG0659-SRM4)					Prepared & Analyzed: 7/21/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG0710										
Blank (BEG0710-BLK1)					Prepared: 7/20/2023	Analyzed: 7/27/2023				
Total Filterable Solids (TDS)	ND		10.0	mg/L						
LCS (BEG0710-BS1)					Prepared: 7/20/2023	Analyzed: 7/27/2023				
Total Filterable Solids (TDS)	28.8		10.0	mg/L	2000		1.44	0-200		
Duplicate (BEG0710-DUP1)		**Source: 23G1435-01**			Prepared: 7/20/2023	Analyzed: 7/27/2023				
Total Filterable Solids (TDS)	2860		10.0	mg/L		2820			1.41	10
Duplicate (BEG0710-DUP2)		**Source: 23G1456-01**			Prepared: 7/20/2023	Analyzed: 7/27/2023				
Total Filterable Solids (TDS)	1030		10.0	mg/L		1040			0.643	10
Reference (BEG0710-SRM1)					Prepared: 7/20/2023	Analyzed: 7/27/2023				
Total Filterable Solids (TDS)	333			mg/L	325.0		103	90-110		
Reference (BEG0710-SRM2)					Prepared: 7/20/2023	Analyzed: 7/27/2023				
Total Filterable Solids (TDS)	517			mg/L	495.0		104	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG0809										
Blank (BEG0809-BLK1)					Prepared: 7/21/2023	Analyzed: 7/24/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
Blank (BEG0809-BLK2)					Prepared: 7/21/2023	Analyzed: 7/24/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
LCS (BEG0809-BS1)					Prepared: 7/21/2023	Analyzed: 7/24/2023				
Kjeldahl Nitrogen (TKN), Total	5.57		1.00	mg/L	5.709		97.5	90-110		
LCS (BEG0809-BS2)					Prepared: 7/21/2023	Analyzed: 7/24/2023				
Kjeldahl Nitrogen (TKN), Total	5.31		1.00	mg/L	5.709		93.0	90-110		
Duplicate (BEG0809-DUP1)		**Source: 23G1451-01**			Prepared: 7/21/2023	Analyzed: 7/24/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Duplicate (BEG0809-DUP2)		**Source: 23G1471-01**			Prepared: 7/21/2023	Analyzed: 7/24/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Matrix Spike (BEG0809-MS1)		**Source: 23G1451-01**			Prepared: 7/21/2023	Analyzed: 7/24/2023				
Kjeldahl Nitrogen (TKN), Total	9.32		1.40	mg/L	7.992	ND	117	90-110		
Matrix Spike (BEG0809-MS2)		**Source: 23G1471-01**			Prepared: 7/21/2023	Analyzed: 7/24/2023				
Kjeldahl Nitrogen (TKN), Total	8.23		1.40	mg/L	7.992	ND	103	90-110		
Reference (BEG0809-SRM1)					Prepared: 7/21/2023	Analyzed: 7/24/2023				
Kjeldahl Nitrogen (TKN), Total	23.4			mg/L	23.80		98.4	90-110		

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Ranch: Rafael Pacheco

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG0838										
Blank (BEG0838-BLK1)					Prepared: 7/24/2023	Analyzed: 7/26/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.5		1.0	units						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEG0838-BLK2)					Prepared: 7/24/2023	Analyzed: 7/26/2023				
pH	5.9		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEG0838-BLK3)					Prepared: 7/24/2023	Analyzed: 7/26/2023				
pH	6.0		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Duplicate (BEG0838-DUP1)		**Source: 23G1468-01**			Prepared: 7/24/2023	Analyzed: 7/26/2023				
pH	7.6		1.0	units		7.6			0.264	10
Electrical Conductivity	2.35		0.01	mmhos/cm		2.37			0.729	10
Electrical Conductivity umhos	2350		10.0	umhos/cm		2370			0.729	10
Duplicate (BEG0838-DUP2)		**Source: 23G1476-06**			Prepared: 7/24/2023	Analyzed: 7/26/2023				
Electrical Conductivity	0.96		0.01	mmhos/cm		0.97			0.894	10
pH	7.7		1.0	units		7.7			0.260	10
Electrical Conductivity umhos	958		10.0	umhos/cm		966			0.894	10
Reference (BEG0838-SRM1)					Prepared: 7/24/2023	Analyzed: 7/26/2023				
Electrical Conductivity	525			umhos/cm	538.0		97.5	90-110		
Reference (BEG0838-SRM2)					Prepared: 7/24/2023	Analyzed: 7/26/2023				
pH	7.8			units	7.790		100	.7163-101.28		
Reference (BEG0838-SRM3)					Prepared: 7/24/2023	Analyzed: 7/26/2023				
Electrical Conductivity	975			umhos/cm	1000		97.5	90-110		
Reference (BEG0838-SRM4)					Prepared: 7/24/2023	Analyzed: 7/26/2023				
Electrical Conductivity	974			umhos/cm	1000		97.4	90-110		
Reference (BEG0838-SRM5)					Prepared: 7/24/2023	Analyzed: 7/26/2023				
Electrical Conductivity	982			umhos/cm	1000		98.2	90-110		
Reference (BEG0838-SRM6)					Prepared: 7/24/2023	Analyzed: 7/26/2023				

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Rafael Pacheco

Received: 07/20/2023 15:31
Reported: 07/31/2023 08:53

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG0838 (Continued)										
Reference (BEG0838-SRM6)					Prepared: 7/24/2023	Analyzed: 7/26/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEG0838-SRM7)					Prepared: 7/24/2023	Analyzed: 7/26/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEG0838-SRM8)					Prepared: 7/24/2023	Analyzed: 7/26/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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07/20/23 15:31 23G1468

JG

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples 1 No. of Bottles:

Purchase Order No | Bill To: | Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State: CA** **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:

REQUESTED BY: Rui & John Avila
PROJECT Rafael Pacheco
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

( ) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
(X) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	Gear Head Well	7/19/2023	1600	-0.6	
2					
3					
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	7/19/2023 @ 1600	7/20/23 @ 900
Second	[illegible]	Dellavalle Laboratory, Inc.	7/20/23@ 900	7/20
Third				
Fourth	[illegible]	DCL	7/20/23 (5:3)	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

07/20/23 15:31 23G1468

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice: ☒ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)	1									
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1									
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										

pH Strips
Lot 10BDH4501
Exp. Jan. 2025

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Rafael Pacheco Dairy

Received: 11/09/2023 16:20
Reported: 11/16/2023 10:31

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0480-01	Well A House	Ag Water	T. Madruga		11/08/2023 14:00
23K0480-02	Well B Barn	Ag Water	T. Madruga		11/08/2023 14:30

Default Cooler Temperature on Receipt °C: 0.4
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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LABORATORY INC

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8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Rafael Pacheco Dairy

Received: 11/09/2023 16:20
Reported: 11/16/2023 10:31

Sample Results

Sample: Well A House
23K0480-01 (Water)

Sampled: 11/8/2023 14:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.53**	mmhos/cm	0.01	1		11/15/23 11:11	SM 2510 B		BEK0350
Electrical Conductivity umhos	**532**	umhos/cm	10.0	1		11/15/23 11:11	SM 2510 B		BEK0350
Ammonia (as N)	ND	mg/L	0.00	1		11/08/23 14:00	Field		BEK0302
Nitrate Nitrogen as NO3N	**1.2**	mg/L	0.1	1	10	11/09/23 19:43	EPA 300.0		BEK0247
Temperature	**25.0**	units	0.0	1		11/15/23 11:11	SM 4500-H+	H	BEK0350
pH	**7.9**	units	1.0	1		11/15/23 11:11	SM 4500-H+	H	BEK0350

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Rafael Pacheco Dairy

Received: 11/09/2023 16:20
Reported: 11/16/2023 10:31

Sample Results
(Continued)

Sample: Well B Barn
23K0480-02 (Water)

Sampled: 11/8/2023 14:30
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.52**	mmhos/cm	0.01	1		11/15/23 11:13	SM 2510 B		BEK0350
Electrical Conductivity umhos	**516**	umhos/cm	10.0	1		11/15/23 11:13	SM 2510 B		BEK0350
Ammonia (as N)	ND	mg/L	0.00	1		11/08/23 14:30	Field		BEK0302
Nitrate Nitrogen as NO3N	**1.1**	mg/L	0.1	1	10	11/09/23 20:03	EPA 300.0		BEK0247
Temperature	**25.0**	units	0.0	1		11/15/23 11:13	SM 4500-H+	H	BEK0350
pH	**8.0**	units	1.0	1		11/15/23 11:13	SM 4500-H+	H	BEK0350

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Rafael Pacheco Dairy

Received: 11/09/2023 16:20
Reported: 11/16/2023 10:31

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0247										
Blank (BEK0247-BLK1)					Prepared & Analyzed: 11/9/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0247-BLK2)					Prepared & Analyzed: 11/9/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0247-BLK3)					Prepared: 11/9/2023 Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0247-BS1)					Prepared & Analyzed: 11/9/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		101	90-110		
LCS (BEK0247-BS2)					Prepared: 11/9/2023 Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.5	90-110		
Duplicate (BEK0247-DUP1)		**Source: 23K0409-01**			Prepared & Analyzed: 11/9/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L		ND				10
Duplicate (BEK0247-DUP2)		**Source: 23K0487-01**			Prepared: 11/9/2023 Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	2.1		0.1	mg/L		2.1			1.25	10
Matrix Spike (BEK0247-MS1)		**Source: 23K0409-01**			Prepared & Analyzed: 11/9/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000	ND	97.5	90-110		
Matrix Spike (BEK0247-MS2)		**Source: 23K0487-01**			Prepared: 11/9/2023 Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	6.9		0.1	mg/L	5.000	2.1	97.2	90-110		
Reference (BEK0247-SRM1)					Prepared & Analyzed: 11/9/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		99.0	90-110		
Reference (BEK0247-SRM2)					Prepared & Analyzed: 11/9/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		98.6	90-110		
Reference (BEK0247-SRM3)					Prepared: 11/9/2023 Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		99.1	90-110		

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DELLAVALLE™ LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Rafael Pacheco Dairy

Received: 11/09/2023 16:20
Reported: 11/16/2023 10:31

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0350										
Blank (BEK0350-BLK1)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.6		1.0	units						
Blank (BEK0350-BLK2)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.6		1.0	units						
Blank (BEK0350-BLK3)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.6		1.0	units						
Duplicate (BEK0350-DUP1)		**Source: 23K0483-01**			Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	3.23		0.01	mmhos/cm		3.27			1.28	10
pH	7.4		1.0	units		7.4			0.269	10
Electrical Conductivity umhos	3230		10.0	umhos/cm		3270			1.28	10
Duplicate (BEK0350-DUP2)		**Source: 23K0518-04**			Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	0.68		0.01	mmhos/cm		0.68			0.337	10
Electrical Conductivity umhos	685		10.0	umhos/cm		682			0.337	10
pH	7.8		1.0	units		7.8			0.256	10
Reference (BEK0350-SRM1)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	576			umhos/cm	538.0		107	90-110		
Reference (BEK0350-SRM2)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEK0350-SRM3)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	1080			umhos/cm	1000		108	90-110		
Electrical Conductivity umhos	1080			umhos/cm	1000		108	90-110		
Reference (BEK0350-SRM4)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	1080			umhos/cm	1000		108	90-110		
Electrical Conductivity umhos	1080			umhos/cm	1000		108	90-110		
Reference (BEK0350-SRM5)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	1090			umhos/cm	1000		109	90-110		

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Rafael Pacheco Dairy

Received: 11/09/2023 16:20
Reported: 11/16/2023 10:31

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0350 (Continued)										
Reference (BEK0350-SRM5)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity umhos	1090			umhos/cm	1000		109	90-110		
Reference (BEK0350-SRM6)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0350-SRM7)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0350-SRM8)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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11/09/23 16:20 23K0480

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

SG

No. Samples: 2 No. of Bottles:

Purchase Order No

Bill To: Acct # 19840 Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com

COPY TO:

REQUESTED BY: Rui & John Avila
PROJECT Rafael Pacheco Dairy
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents

Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	Well A House	11/8/2023	1400	0.4	ND
2	Well B Barn	11/8/2023	1430	0.9	ND
3					
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/8/2023 @ 1430	11/9/2023 @9am
Second	JLF	Dellavalle Laboratory, Inc.	11/9/2023 @9am	11-9-23 3:55
Third				
Fourth			11/9 16:20	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:
Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/09/23 16:20 23K0480

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☑ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☑ None ☐ Box ☐

Refrigerant: Wet Ice:☑ Blue Ice:☐ None:☐

Container(s) Received	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1								
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										